

ContractAI

AI Contract Review & Management Platform

System Design Document

Version 1.0 | August 2026

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Status	Draft
Classification	Confidential
Last Updated	August 8, 2026

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1. Executive Summary

ContractAI is an AI-powered contract review and lifecycle management platform designed for Indian SMBs and startups. The platform eliminates the need for expensive legal counsel for routine contract review by using large language models to analyze contract documents, identify risky clauses, and suggest modifications.

Indian startups and SMBs routinely sign contracts with significantly larger counterparties (enterprise clients, investors, vendors) where the power dynamic results in one-sided agreements. ContractAI levels the playing field by providing instant, affordable contract intelligence that was previously available only to companies with in-house legal teams.

Key capabilities include:

- Upload any contract (PDF or DOCX) and receive a comprehensive risk analysis within minutes
- Automatic detection of risky clauses: unlimited liability, broad IP assignment, one-sided termination, auto-renewal traps, and aggressive non-compete provisions
- Plain-language explanations of legal jargon so non-lawyers can understand what they are signing
- AI-generated counter-proposals with tracked changes ready for negotiation
- Contract template library with pre-vetted templates for common Indian business agreements
- Full lifecycle management with version control, e-signature, and deadline tracking
- Indian law compliance checks covering the Indian Contract Act, IT Act, and DPDP Act

The platform targets the estimated 1.4 million active startups and 63 million MSMEs in India that collectively sign millions of contracts annually with little or no legal review.

2. System Architecture

ContractAI combines document processing, vector search, and LLM-powered analysis in a pipeline architecture optimized for accuracy and speed.

2.1 Technology Stack

Layer	Technology
Backend API	FastAPI (Python 3.12) with async processing
Frontend	React / Next.js 14 with TypeScript
Primary Database	PostgreSQL 16 with full-text search
Vector Store	Qdrant for clause similarity search
Cache & Queue	Redis 7 (BullMQ for document processing pipeline)
AI / LLM	OpenRouter (GPT-4o-mini, Claude Haiku) for clause analysis
Document Parsing	PyMuPDF (PDF), python-docx (DOCX), OCR via Tesseract
E-Signatures	Digio API (India-specific, Aadhaar e-sign)
File Storage	S3-compatible object storage with client-side encryption

2.2 Document Processing Pipeline

When a user uploads a contract, the system processes it through a multi-stage pipeline:

- Stage 1 - Ingestion: File validation, virus scan, format detection (PDF/DOCX), and storage
- Stage 2 - Extraction: Text extraction with layout preservation. OCR for scanned documents. Tables, headers, and clause boundaries are detected.
- Stage 3 - Clause Segmentation: The document is split into individual clauses using heading detection, numbering patterns, and ML-based boundary detection.
- Stage 4 - Embedding: Each clause is embedded using a sentence transformer model and stored in Qdrant for similarity search against known clause patterns.
- Stage 5 - Risk Analysis: Each clause is analyzed by the LLM against a risk taxonomy. Known patterns are matched via vector search; novel clauses get full LLM review.
- Stage 6 - Report Generation: Risk scores, explanations, and suggestions are compiled into a structured review report.

2.3 Vector Search Architecture

ContractAI maintains a curated vector database of clause patterns organized by category (liability, IP, termination, payment, confidentiality, etc.). When a new clause is analyzed, the system first searches for similar clauses in the vector store. If a high-confidence match is found, the pre-computed risk assessment is used directly, reducing LLM API calls and improving consistency. Novel or ambiguous clauses are sent to the LLM for full analysis, and the result is added to the vector store for future reference.

3. Data Model

The database schema supports the full contract lifecycle from upload through signature and renewal tracking. All tables include `organization_id` for multi-tenant isolation.

3.1 Core Tables

Table	Purpose	Key Fields
organizations	Tenant companies	name, domain, plan_tier, settings
contracts	Contract documents	title, type, status, counterparty_id, effective_date
clauses	Extracted clause segments	contract_id, text, category, position, risk_level
risk_assessments	AI risk analysis results	clause_id, risk_score, explanation, suggestion
suggestions	AI-generated modifications	clause_id, original_text, suggested_text, rationale
templates	Contract templates	name, type, content, variables, version
template_variables	Template placeholder fields	template_id, key, label, type, default_value
versions	Contract version history	contract_id, version_no, file_url, changes_summary
signatures	E-signature records	contract_id, signer_name, signer_email, signed_at
counterparties	External parties	name, email, company, contracts_count
tags	Contract labels and categories	name, color, org_id
comments	Review comments and discussions	contract_id, clause_id, user_id, text, resolved

3.2 Key Relationships

- contracts belong to organizations and reference a counterparty
- clauses are extracted from contracts; each clause has one `risk_assessment` and zero or more `suggestions`
- versions track the full history of contract modifications
- signatures are collected per contract with timestamp and verification
- comments can be attached to specific clauses for collaborative review
- templates contain `template_variables` that are substituted during contract generation

4. Core Features

4.1 AI Contract Review

Users upload any contract in PDF or DOCX format and receive a comprehensive review within 2-5 minutes. The review includes:

- Overall risk score (0-100) with risk level classification
- Per-clause risk assessment with plain-language explanations
- Specific suggestions for modifications to reduce risk
- Comparison against market-standard terms for the contract type
- Indian law compliance flags for Contract Act, IT Act, and DPDP requirements

4.2 Risky Clause Detection

The system is trained to identify the following categories of risky clauses:

Clause Type	Risk Level	Example Flags
Unlimited Liability	Critical	No cap on damages; indemnification without limits
Broad IP Assignment	Critical	All work product IP transfers to client including background IP
One-sided Termination	High	Client can terminate at will; vendor needs 90-day notice
Auto-Renewal	Medium	Contract auto-renews without explicit opt-in; short cancellation window
Non-Compete	High	Broad geographic/industry scope; duration exceeds 12 months
Exclusivity	High	Prevents working with competitors; no reciprocal obligation
Data Rights	Medium	Broad data usage rights; no deletion obligation post-termination
Payment Terms	Medium	Payment beyond Net-60; no late payment penalties for client
Governing Law	Low	Foreign jurisdiction; arbitration in inconvenient location

4.3 Plain-Language Explanations

Every risky clause is accompanied by a plain-language explanation that a non-lawyer can understand. For example, a clause stating 'The Service Provider shall indemnify and hold harmless the Client from any and all claims, damages, losses, costs and expenses' is explained as: 'This means you would be financially responsible for ALL problems that arise, even if the client caused them. This is very one-sided.'

4.4 Counter-Proposal Generation

For each risky clause, ContractAI generates a counter-proposal that protects the user while remaining commercially reasonable. Counter-proposals are presented as tracked changes that can be exported as a redlined DOCX for direct negotiation with the counterparty.

4.5 Contract Template Library

The platform includes pre-vetted templates for common Indian business agreements:

- Non-Disclosure Agreement (mutual and one-way)
- Master Service Agreement (IT services, consulting)
- Employment Agreement (full-time, contractor, intern)
- Vendor/Supplier Agreement
- SaaS Subscription Agreement
- Shareholder Agreement
- Advisory Agreement with ESOP provisions
- Freelancer/Contractor Agreement

4.6 Additional Features

- Version history with visual diff comparison between contract iterations
- E-signature integration via Digiio (supports Aadhaar e-sign for India)
- Deadline and renewal tracking with email/WhatsApp alerts 30, 15, and 7 days before key dates
- Clause library for reusing approved language across contracts
- Multi-party review workflow with comments, approvals, and audit trail
- Indian law compliance checks covering Contract Act 1872, IT Act 2000, and DPDP Act 2023

5. Risk Scoring Model

ContractAI uses a multi-factor risk scoring model that produces both an overall contract risk score and individual clause-level scores:

5.1 Scoring Methodology

Score Range	Risk Level	Description
0-25	Low Risk	Standard market terms. Safe to sign with minor review.
26-50	Medium Risk	Some one-sided terms. Review recommended; negotiate key points.
51-75	High Risk	Significantly one-sided. Strong negotiation or legal review advised.
76-100	Critical Risk	Dangerous terms present. Do not sign without legal counsel.

5.2 Scoring Factors

- Clause severity: weighted by clause category (liability > payment terms > governing law)
- Reciprocity: one-sided obligations increase score; mutual obligations are neutral
- Market comparison: deviation from standard terms for the contract type and industry
- Compliance: non-compliance with Indian law requirements adds mandatory risk points
- Cumulative risk: multiple medium-risk clauses can elevate overall score to high risk

6. API Design

ContractAI provides a RESTful API for all operations, with JWT authentication and optional API key access for enterprise integrations.

6.1 Contract Endpoints

Method	Endpoint	Description
POST	/contracts/upload	Upload contract for review
GET	/contracts/:id/review	Get risk analysis results
POST	/contracts/:id/counter-proposal	Generate counter-proposal
GET	/contracts/:id/versions	List version history
POST	/contracts/:id/sign	Initiate e-signature flow

6.2 Template & Alert Endpoints

Method	Endpoint	Description
GET	/templates	List available templates
POST	/templates/:id/generate	Generate contract from template
GET	/alerts/upcoming	Get upcoming deadlines and renewals

POST	/contracts/:id/comments	Add review comment
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7. Revenue Model

ContractAI offers both per-review pricing for occasional users and monthly subscriptions for regular users:

Plan	Price	Features
Pay-per-Review	\$29-99/review	Single contract review, risk report, suggestions. Price varies by length.
Starter	\$149/mo	10 reviews/month, template library, deadline tracking, 3 users
Professional	\$299/mo	30 reviews/month, counter-proposals, e-signatures, clause library, 10 users
Enterprise	\$499/mo	Unlimited reviews, custom templates, API access, SSO, dedicated support

8. Security

Contract documents contain some of the most sensitive business information. ContractAI implements defense-in-depth security:

- End-to-end encryption: Documents encrypted at rest (AES-256-GCM) with per-organization keys and in transit (TLS 1.3)
- Zero-retention LLM policy: Contract text sent to LLMs is not retained by the AI provider; we use API plans with zero data retention
- Access control: Role-based access (Admin, Reviewer, Viewer) with per-contract permission overrides
- Audit trail: Every access, review, modification, and download is logged with timestamp, user, and IP
- Data retention: Configurable retention policies; automatic deletion after contract expiry + retention period
- SOC 2 Type II compliance roadmap for enterprise customers
- Penetration testing: Quarterly third-party security assessments
- Data residency: All data stored in India-region servers for DPDP compliance

9. Testing Strategy

ContractAI requires rigorous testing due to the legal nature of its output:

9.1 Test Categories

Category	Target	Description
Clause Extraction	> 95% accuracy	Correct identification and segmentation of

		individual clauses
Risk Classification	> 90% precision	Risk level matches expert legal assessment
Suggestion Quality	> 85% acceptance	User acceptance rate of AI-generated suggestions
Template Generation	100% valid	Generated contracts are syntactically and legally valid
OCR Accuracy	> 98% character	Text extraction from scanned documents

9.2 Testing Approach

- Golden dataset: 500+ contracts manually reviewed by lawyers, used as ground truth for accuracy testing
- Regression suite: Every new model version is tested against the golden dataset before deployment
- A/B testing: New suggestion algorithms are tested against existing ones on real user interactions
- Integration tests: End-to-end pipeline tests using testcontainers (PostgreSQL, Redis, Qdrant)
- Security tests: OWASP-based security testing for document upload, API access, and data leakage

10. Milestones

ContractAI follows a 13-week timeline to general availability:

Week	Phase	Deliverables
1-2	Foundation	Document parsing pipeline, clause extraction, basic UI for upload
3-5	MVP	Risk analysis engine, risk report generation, dashboard, per-review billing
6-7	Intelligence	Counter-proposal generation, clause library, vector search optimization
8-9	Beta	Template engine, version control, multi-party review, beta user onboarding
10-11	Integration	Digio e-signatures, deadline alerts, Indian law compliance checks
12-13	GA Launch	Production hardening, security audit, documentation, marketing site

Post-GA roadmap includes Hindi language support, bulk contract analysis, contract analytics dashboard, and integration with common Indian business tools (Zoho, Tally).